Analytics in a New Sports Age

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1 Introduction

As basketball has reached new heights in popularity around the world, so has the desire to win and the need to have an edge on other teams have also increased. Sports used to just be about averages and had never anything more complicated beyond that. However, now advanced analytics have begun taking a stronger influence in how basketball is played. In this article, I will talk about some of the analytical models used in the National Basketball Association to evaluate individual players in their each team's game strategies.

2 Basketball

2.1 Basic Statistics

The most commonly quoted statistics quoted by casual fans are points per game, assists per game, and rebounds per game, which are all averages. Accuracy is measured by field goal percentage and free throw percentage, which are both percentages of makes over attempts. These statistics can be misleading as they don't tell the whole story of the value of a player to their team.

2.2 Player Efficiency Rating

Player Efficiency Rating, better known as PER, is a statistic created by John Hollinger to measure a player's per minute productivity. The unadjusted value is calculated in the following formula. Most of the

```
uPER = (1 / MP) *
     [ 3P
     + (2/3) * AST
     + (2 - factor * (team_AST / team_FG)) * FG
     + (FT *0.5 * (1 + (1 - (team_AST / team_FG)) + (2/3) * (team_AST / team_FG)))
     - VOP * TOV
     - VOP * DRB% * (FGA - FG)
     - VOP * 0.44 * (0.44 + (0.56 * DRB%)) * (FTA - FT)
     + VOP * (1 - DRB%) * (TRB - ORB)
     + VOP * DRB% * ORB
     + VOP * STL
     + VOP * DRB% * BLK
     - PF * ((lg_FT / lg_PF) - 0.44 * (lg_FTA / lg_PF) * VOP) ]
```

Figure 1: PER formula

```
factor = (2 / 3) - (0.5 * (lg_AST / lg_FG)) / (2 * (lg_FG / lg_FT))
VOP    = lg_PTS / (lg_FGA - lg_ORB + lg_TOV + 0.44 * lg_FTA)
DRB%   = (lg_TRB - lg_ORB) / lg_TRB
```

Figure 2: Meanings of terms

terms are contractions for basic statistics used in the league, like assists and minutes played. Some more definitions (by order of appearance):

- team = player's team
- VOP = value of possession
- DRB% = defensive rebound percentage
- = lg = league

The 1/MP (1/Minutes Played) is used to scale the PER based on minutes played. The player's positive contributions, like three pointers and assists, are added while the negative contributions, like turnovers and personal fouls, are subtracted.

The unadjusted PER is then adjusted by multiplying a pace adjustment, which is league pace divided by team pace. Since different teams play at different paces, a player's production also changes from team to team, thus the need for a pace adjustment. This is to scale player's PER to show a player's contribution to the team rather than just a result of the team's system of playing.

The adjusted PER is finally multiplied by (15/league average PER) to set the league average PER to 15 for better comprehension by fans. A PER above 15 means better contributions than an average player while below 15 means worse.

2.3 Win Shares

Win shares are an estimation of the amount of wins a player contributed to their team's season. Win shares are a sum of offensive win shares and defensive win shares.

Offensive win shares are calculated by marginal offense divided by marginal points per win and defensive win shares are calculated by marginal defense divided by marginal points per win, so win shares are calculated by

$$\frac{\text{marginal offense} + \text{marginal defense}}{\text{marginal points per win}}.$$

Marginal offense is calculated by the formula

$$\text{points produced} - (0.92)(\text{league points per possession})(\text{offensive possessions}).$$

This is essentially the difference in points produced by the player and by an average player. Marginal defense is calculated by the formula:

$$(1.08)\left(\frac{\text{player minutes}}{\text{team minutes}}\right)(\text{team defensive possessions})\left(\text{league points per possession} - \frac{\text{defensive rating}}{100}\right).$$

This is essentially the difference in points allowed by an average player and the player. Marginal points per win is calculated by the formula

$$(0.32)(\text{league points per game})\left(\frac{\text{team pace}}{\text{league pace}}\right).$$

This is essentially the factor to standardize win shares for each team.

The sum of the win shares of all the members of a team should sum to the number of wins the team had. There is an average margin of error of 2.74 wins.

2.4 Conclusion

Sports, like basketball are becoming more ingrained with math and statistics. The two examples that I presented are a few of the many different analytics that a team uses to evaluate its players. If a player displays an above average PER and positive win shares, they are deemed conducive to winning basketball. These statistics can also be used to compare with other players from other teams to influence transactions in the league. Since the NBA is a multi-billion dollar organization, these transactions can be worth millions of dollars to teams. This real-world application of statistics and math show us its value.

References

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